

(Evaporation)

Pre-Lab Plan

Unit Operations Lab 1

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Group 1, Tuesday, AM section

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1. Experimental Objective (5 pts)

Determine how absolute pressure affects the performance of a single-stage evaporator concentrating a 10 wt% sugar solution, with all runs at constant solution concentration and steady state. For this purpose, we will quantify capacity and steam economy at three vacuum levels and evaluate whether the observed trends align with evaporation and heat-transfer theory, or more specifically the relationships between pressure, boiling temperature, overall heat-transfer coefficient (UA), and the driving temperature difference (ΔT), while also checking the overall energy balance.

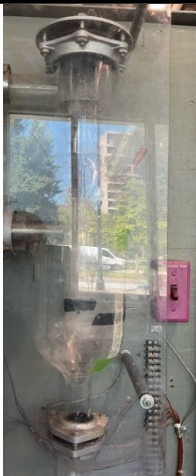
2. Experimental

2.1 Apparatus (5 pts)

Several pieces of equipment are used in the single-stage evaporator setup, but the main apparatus include the vacuum pump, heat exchanger, condenser, and glass vessel. The complete list of equipment, along with images and descriptions, is provided in Table 1.

Equipment	Number	Picture	Description
Condenser	1		A device that condenses the steam boiled out of the sugar-water solution

Pressure Gauge	2		A device that measures and displays the pressure being delivered to the vessel.
Vacuum Pump	3		A device used to reduce the pressure of the vessel, which lowers the boiling point.
Heat Exchanger	4		A piece of equipment that receives steam and heats up the sugar-water solution and brings it to boiling, which separates the sugar and water in the vessel, allowing the steam to be transferred into the condenser.

Steam Trap	5		A piece of equipment that condenses the steam fed from the wall into the heat exchanger, which is then transferred to the steam condensate flask.
Flowmeter	6		A piece of equipment used to measure and adjust the flow rate of cooling water,
Glass Vessel	7		A piece of equipment used to store the sugar-water solution and isolate the sugar after boiling

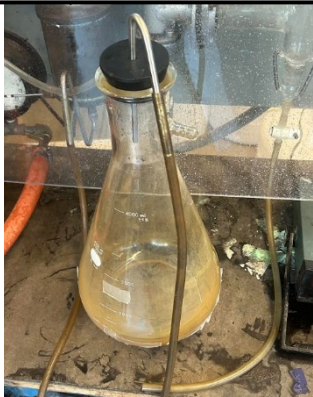
Thermocouple	8		A device used to measure the inlet steam temperature
Steam condensate flask (4L)	9		Collects the steam that is condensed using the steam trap
Water condensate flasks x2 (2L)	 		One of the flasks will be used to collect the water condensate, and the other is used to ensure steady flow before collecting data.

Table 1: Detail Apparatus List

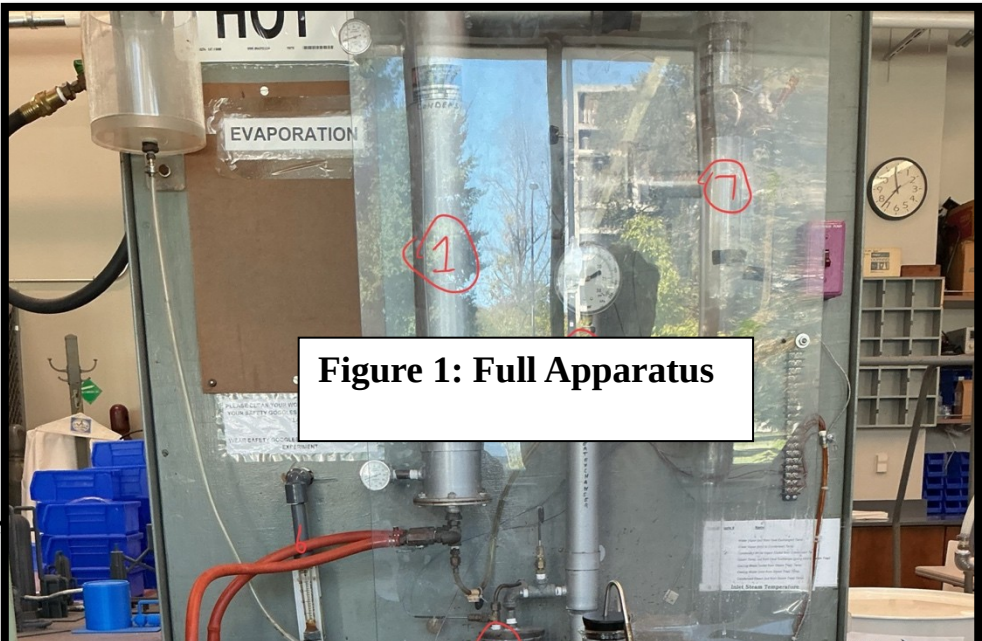
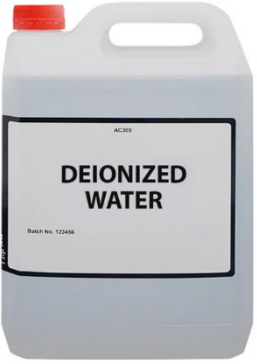
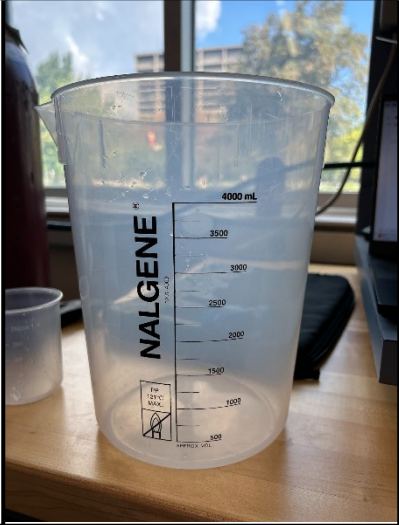
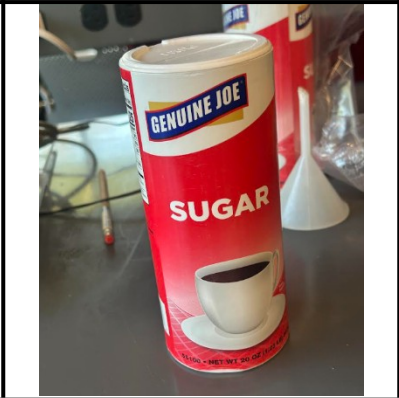


Figure 1: Full Apparatus

2.2 Materials and Supplies (5 pts)

The complete material and supply list required for operating the single-stage evaporator are provided in Table 2. This table includes all consumables and other resources needed for the experiment, along with relevant details for each item.

Materials/ supplies	Picture	Description
100 mL graduated cylinders		Used to measure and transfer precise liquid volumes
Deionized water		Used to ensure stable levels in the glass vessel

Beakers (1-4 L capacity)		Containers for mixing and preparing solutions
Stirring rods		For manual mixing support during solution preparation
Genuine Joe Sugar		used as a solute for preparing solutions in the experiment

Orange heat-resistant gloves		Provides hand protection while handling hot or hazardous materials
White plastic funnel		Used for transferring liquids safely into narrow containers
Adjustable metal scissor jack platform		used to support and adjust the height of lab equipment such as beakers, flasks, or heating elements

Table 2: Material and Supply List

2.3 Experimental Plan (10 pts)

Solution Preparation:

Weigh out 300 g of sucrose and dissolve it in about 2700 g (≈ 2700 mL) of deionized water to make 3 L of a 10 wt% sugar solution. Stir until completely dissolved.

Experimental Procedure

- 1) Drain separator if needed. Fill with 10 wt% sugar solution to the level line.
- 2) Add deionized water to the makeup tank. Keep sugar solution level constant.
- 3) Start steam flow and the steam trap cooling water.
- 4) Use the vacuum pump to set the first pressure. Wait until the solution boils and circulates through the heat exchanger.
- 5) At steady state, empty condensate flasks. Collect steam condensate and water product for 5–10 min. Record volumes and calculate flow rates.
- 6) Record cooling water flow and inlet/outlet temperatures. Use this to calculate steam condensation and energy transfer.
- 7) Repeat measurements at two more vacuum pressures (total of three). Lower pressure reduces boiling point, increases ΔT with steam, and changes capacity vs. economy.

Variables

- Independent variable: absolute pressure in separator (set via vacuum level on the gauge).
- Controlled variables: feed composition (10 wt% sugar), liquid level in separator (maintained at the mark), steam supply valve position, condenser cooling-water flow (held constant), and ambient lab conditions.
- Dependent variables: capacity, economy, ΔT (steam-side condensing temperature – boiling liquid temperature), cooling-water heat duty, and product/steam flow rates.

Data and Model Correlation

- Heat Transfer: Capacity will be compared to the heat-transfer model $q=UA\Delta T$. From measured capacity and heat, UA can be estimated. $\Delta T = T_{\text{steam condensed}} - T_{\text{boiling}}$
- Pressure vs Boiling point: the measured temperatures will be plotted against absolute pressure and compared with steam-table values.
- Capacity ($\dot{m}_{\text{product}}$) and Economy ($\frac{\dot{m}_{\text{product}}}{\dot{m}_{\text{steam}}}$) will be plotted against absolute pressure.
The trends would be compared with theoretical expectations for a single-effect evaporator.
- Energy Balance: heat from steam condensate will be compared with heat duty (evaporation load), and the percent closure will be calculated.

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The vacuum pump and heating equipment require electricity. Water/steam leaks near electrical connections could create a shock risk.
Potential mechanical hazards?	Yes	If yes, identify conditions: Moving parts in the vacuum pump and glass vessel breakage under pressure/vacuum may cause mechanical injury. Handle glassware carefully and avoid over-tightening fittings.
Potential pressure hazards?	Yes	If yes, identify conditions: The evaporator operates under vacuum, and improper sealing may lead to implosion of glass vessels. Steam lines also operate at elevated pressures and can leak if not properly connected.
Potential temperature hazards?	Yes	If yes, identify conditions: Steam, condensers, and heat exchangers operate at elevated temperatures. Hot surfaces and steam lines can cause burns. Condensers and steam traps may release hot condensate.
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Heat-resistant gloves for handling glassware exposed to hot steam or condensate.
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Sucrose (granulated white sugar)- SDS Deionized water- SDS
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	~100 °C	Up to ~100 °C in the steam and solution during boiling. There is a burn risk. To avoid this problem, use insulated gloves and keep face/skin away from hot lines.
Pressure (psia)	~5–15 psia	Operates under vacuum (~5–15 psia). There is a glass vessel implosion risk if mishandled. To avoid this problem, inspect glass for cracks before use, never exceed design limits.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
PPE for operation:	yes	Wear heat-resistant gloves, chemical splash goggles, and clothes that cover the skin during operation to protect against burns from hot surfaces, steam, and splashes from boiling sugar solution.
Pressure control and relief:	yes	Use a pressure regulator and relief valve on the steam line;

